Declaration on the use of AI

Learning and understanding of course material: I used various AI tools to support my study of the course material and understanding of the project assignment. This included requesting intuitive examples to clarify complex concepts, summaries of unfamiliar topics, and questions to help me consolidate my understanding of newly learned concepts.

Code development and implementation: I relied on Claude AI for the implementation of the project in Python, as my programming experience is limited and the time available did not allow me to develop sufficient proficiency independently. I supervised the coding process and used AI assistance as well as lecture materials to understand the code.

Model specification and interpretation of results: I used AI tools as a reasoning aid when considering model specifications, identifying instruments, and interpreting the results obtained from the estimations. In particular, I used them to formulate questions and explore the rationale behind modelling choices, the meaning of the results, and their potential implications with the purpose to support my critical assessment and understanding of the analysis.

Structuring written work: I used AI tools to develop broad outlines and checklists to help organise the paragraphs and ensure that relevant concepts were not overlooked.

English language editing: I used AI for editorial support in identifying grammatical mistakes, improving English phrasing, and supporting in the formulation of more academic-style sentences in individually produced text; proofreading assistance.

Document formatting: assist with converting the project document from Markdown and Python output to LaTeX.